

HAMZA AYOOB

Protection & Control (P&C) Commissioning & Testing Specialist
National Grid Company of Pakistan

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Professional Summary

Testing & Commissioning Engineer with 5+ years of experience in 500kV / 220kV / 132kV EHV grid stations. Specialized in relay testing, CT/PT testing, Auto-Transformer diagnostics, and commissioning of protection schemes. Hands-on expertise with Siemens SIPROTEC 4/5, ABB Relion, Micom P40, and GE Multilin relays using Freja 546, Omicron CMC, Sverker 900 and primary injection sets. Experienced in Distance (21), Differential (87), Overcurrent/Earth Fault (50/51), REF (64), and fault analysis using COMTRADE/SIGRA.

Core Competencies

- Primary & Secondary Injection Testing
- Relay Configuration: Siemens / ABB / GE / Micom
- Protection Functions: 87, 21, 51, 50, 64, 27/59, 25
- Trip Circuit Verification & Logic Checks
- End-to-End Protection Scheme Testing
- Fault Analysis using COMTRADE, SIGRA, Wavewin

EHV Equipment Testing

- Auto-Transformer (500/220kV): IR, TTR, WR, C&DF, Vector Group
- CT Testing: Ratio, Polarity, Burden, Knee Point, Primary Injection
- PT Testing: Ratio, Polarity, Burden, IR, C&DF
- Circuit Breakers (SF₆): Timing, Contact Resistance, Coil Tests,

Software & Testing Tools

- Freja 546 • Omicron CMC 356 • Sverker 900 • DIGSI 4/5 • PCM600 • MiCOM S1 Studio
- GE EnerVista • PowerDB • Easergy Studio

Experience

Assistant Manager (P&C) National Grid Company of Pakistan **Dec-2021-Present**

- Conduct relay testing (primary & secondary injection) for Siemens, ABB, Micom & GE relays.
- Perform Distance (21), Differential (87), O/C–E/F, REF and Voltage protection testing.
- Carry out CT, PT and Auto-Transformer testing: IR, TTR, WR, Burden, C&DF, Polarity and Knee Point.
- Commission protection relays for 500kV & 220kV Transformers and Transmission Lines , verifying trip circuits and all I/O signals.

- Execute scheme testing for Line, Auto-Transformer, Busbar & Feeder protection.
- Analyse fault records using COMTRADE, SIGRA; prepare post-fault engineering reports.
- Replace and commission failed CTs, PTs, and SF₆ breakers ensuring safe re-energization.
- Prepare PowerDB test reports, commissioning documentation, and audit-compliant records.

Assistant Manager Operation IESCO Pakistan

Dec 2019-2021

- Supported testing & commissioning teams during shutdowns, energization, and fault rectification tasks at HV substations.
- Operated switching equipment under PTW & LOTO system, coordinated with field staff for safe operation.
- Conducted routine IR tests, megger tests, cable fault checks, and LV/HV diagnostics.
- Assisted with restoration after transformer, cable, and line faults.

Projects & Achievements

- Led end-to-end protection testing during replacement of failed CT/PT & subsequent recommissioning ensuring accurate polarity, burden, and ratio verification.
- Successfully migrated legacy electromechanical relays to numerical SIPROTEC 5 / ABB Relion platforms, completing setting migration and logic revalidation.
- Performed fault analysis for major grid events, identifying root causes such as CT saturation, incorrect zone settings, and carrier communication failures.
- Conducted protection review and testing after Auto-Transformer faults, validating all differential and REF protections before re-energization.
- Led a project to standardize PowerDB templates for relay testing, improving reporting accuracy and speeding up documentation by 30%.
- Completed commissioning of new protection and control panels including wiring checks, functional tests.

Education

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| • B.Sc. Electrical Engineering (Electrical Power) – (UET) Lahore, | 2014-2018 |
| • Master of Public Administration (MPA) Virtual University of Pakistan | 2022-2024 |

Certifications & Trainings

- PEC Registered Engineer – ELECT/70402
- SAP Asset Management Training
- Electrical Safety & Hazard Awareness

Languages

- Urdu
- English